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Advanced Automation

Housing Prices in Lisbon

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Group 14

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1 Introduction

This project aims to explore, analyze, and model a real-world problem using a structured data-driven methodology. The chosen data set is about house-pricing in Lisbon, which will be used to build *ML* models to predict the house price (full sale price and per m^2), if a house is luxurious and its price band from its characteristics and location. These models can be used to identify over- and under-priced houses and to help business users in choosing a house that matches their preferences.

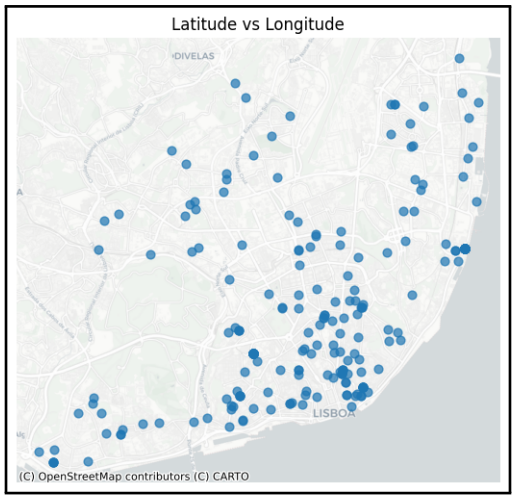


Figure 1: Houses' locations

2 Data Set

Our data set is based on a real-world problem, specifically house pricing from 2021, and is composed of 246 houses with the following characteristics:

Table 1: Dataset variables description

Variable	Description
Id	Identifier of the house in the dataset
Condition	Ranking of the house condition (age/usage)
PropertyType	Type of property (e.g., Home, Single Habitation)
PropertySubType	Apartment, Duplex, Dwelling, Isolated Villa, Penthouse, Studio, Townhouse Dwelling
Bedrooms	Number of bedrooms
Bathrooms	Number of bathrooms
AreaNet	Net usable area
AreaGross	Gross area (including walls and common spaces)
Parking	Number of parking spaces
Latitude	Latitude coordinate
Longitude	Longitude coordinate
Country	Country where the property is located
District	Administrative district
Municipality	Administrative municipality
Parish	Administrative parish
PriceM2	Price per square meter
Price	Total purchase price of the house

3 Data preprocessing and Variable refinement

3.1 Outlier check

An outlier check was performed to verify whether there were any observations that differed significantly from the rest of the dataset, which could lead to problems when the models were to be trained.

The continuous variables Price, Price/m^2, AreaNet and AreaGross were chosen and boxplots were plotted for visual checks. The results are as follows:

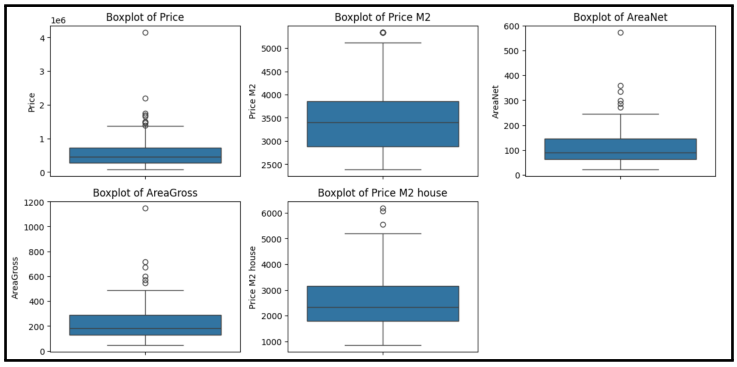


Figure 2: Boxplots for Price and Area

Several outliers stood out, something to be expected when working with data relative to housing. A big capital like Lisbon can accommodate a variety of houses, with very different features, specially in terms like price and area. Like most big capitals, however, asymmetries will prevail, which in turn leads to outliers.

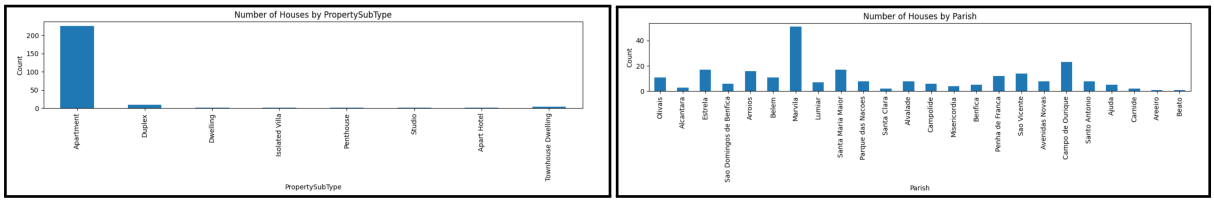
The AreaGross and AreaNet plots, for example, can be easily explained by the fact that big cities favor apartment buildings in detriment of big villas or townhouses. The dataset will consist of most of these smaller houses, but some bigger homes will also appear, leading to said extremities on the plots.

For the Price and Price/m^2, it's noticed that the latter shows less outliers, also in accordance with housing data. Price/m^2 is, by definition, a more normalized variable than simply using Price, such that it's a very common metric used in real estate. Thus, it won't manifest as many outliers, as it removes the area bias that leads to such high prices in some homes.

These outliers can distort model training, specially with linear and logistic regressions and when the dataset isn't particularly large, which is the case for this project.

3.2 Variable count

Following the evident presence of outliers, a count for each variable was carried out, so that it could be evaluated on how they could affect the models. The key results are the following:



(a) Count by PropertySubType **(b)** Count by Parish

Figure 3: Count for PropertySubType and Parish

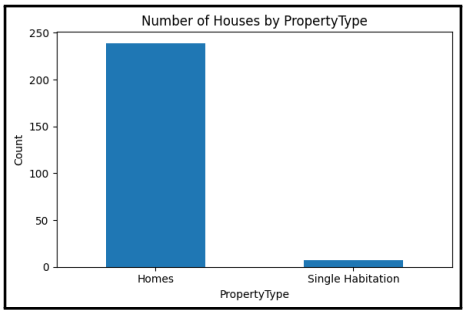


Figure 4: Count by PropertyType

A very high imbalance is quite noticeable in the PropertySubType and PropertyType plots (high number of homes vs Single Habitation and high number of apartments vs other type of houses), which makes sense for a big capital like Lisbon, where, as mentioned before, big apartment buildings are far more common.

Some asymmetries for the houses per Parish are also evident, meaning that some Parishes are more expressive in the dataset, which could lead to some bias, particularly if location effects are a case of study.

In any case, if these variables have a low impact on the target variable, their removal from the test set should be considered to avoid model distortion.

3.3 Location effects

Location is usually one of the most defining factors in housing, and could easily affect the price and/or Price/m^2 of a house, simply by where it's located. This will be studied in Linear Regressions, and to confirm those results an average of Price and Price/m^2 was calculated:

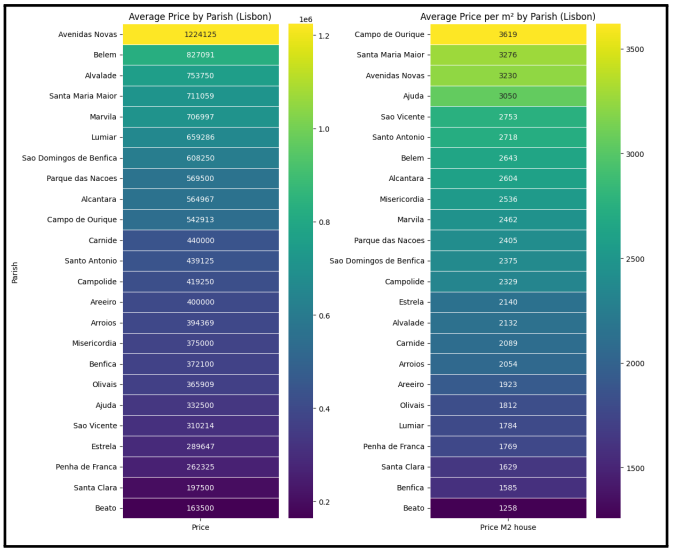
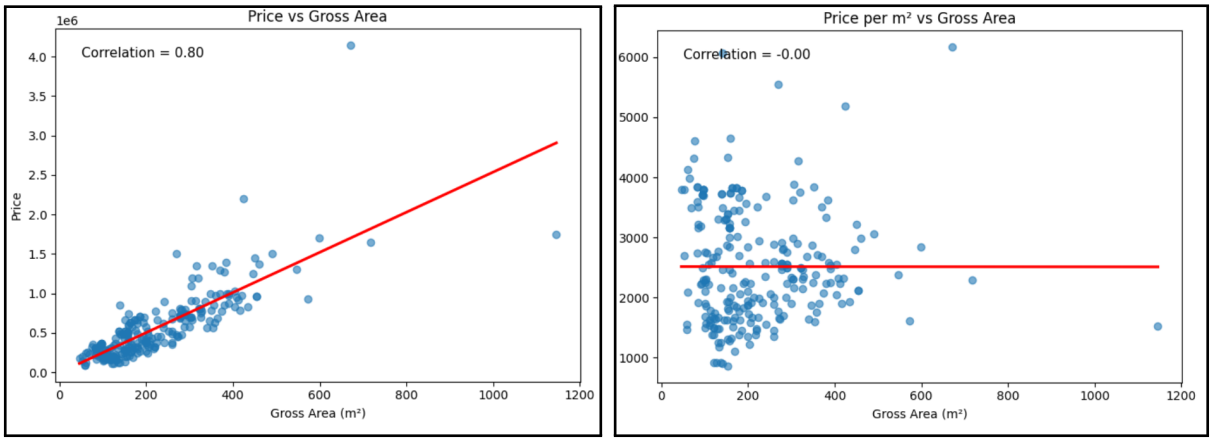


Figure 5: Average Price and Price/m^2 per Parish

It's easy to notice that houses located more towards the center of Lisbon display higher Prices and Price/m^2. As before, this is coincident with data from a big capital, where the prices decrease as you move away from the city center.

3.4 Bivariate analysis

Some variables in the dataset are expected to show some linear correlation, which could prove useful to explain some behaviors that the model can express. Particularly, **Price** and **Gross Area** come to mind, as it makes sense to presume that a house with a larger area displays higher market value. For this reason, **Price** and **Price/m^2** were both plotted against **Gross Area**, giving the below plots:



(a) Price vs Gross Area **(b)** Price/m^2 vs Gross Area

Figure 6: Price and Price/m^2 vs Gross Area

The results show a clear correlation between Price and Gross Area, but no relation between Price/m^2 when plotted against the same variable. This, coupled with the average Price/m^2 obtained in the last subsection, illustrates how Price/m^2 is easily distinguished by location, but much harder to predict with other variables in consideration, like Gross Area, even though it

clearly inflates the price of the houses. This leads to the belief that Price/m^2 will be harder to predict in a simple linear regression, as it will be evaluated later on.

3.5 Data cleansing

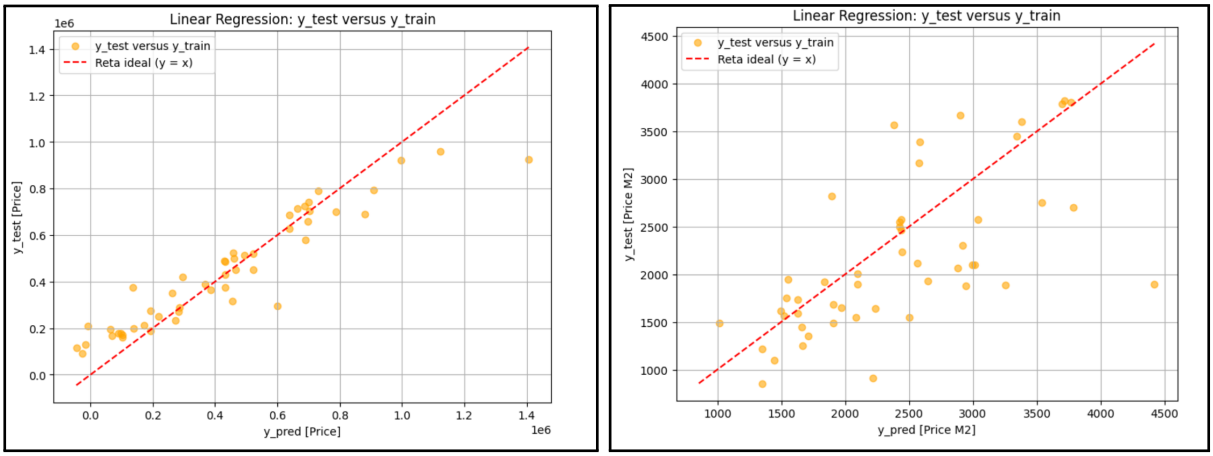
Since all houses are located in Portugal, Lisbon, classes **Country**, **District**, **Municipality**, are redundant, also **Id**, being only the "name" of the house it gives no relevant information, thus being all unnecessary for the model.

A function was created that gave the option to create the models with and without **PropertyType** and **PropertySubType** in the dataset (the two features that the data exploration flagged as being probable to distort the models). The results displayed in this report were obtained without these features, but comparisons with the alternative are mentioned, so that some conclusions could be grasped.

4 Linear Regression

4.1 Model Prediction for Price and Price/m^2

Taking into consideration what the data exploration returned, a linear regression was implemented to try and predict Price and Price/m^2 based on the house characteristics. The results are as follows:



(a) Real Price vs Predicted Price

(b) Real Price/m^2 vs Predicted Price/m^2

Figure 7: Predictions for Price and Price/m^2

4.1.1 Model Evaluation

The results are quite conclusive. The model performed much better predicting price than its m^2 counterpart, something to be expected. Considering the poor correlation between Price/m^2 and Gross Area, it's clear that this variable shows poor correlations with other variables in the dataset, with Parish being an exception, as showed before, leading to a more complex pattern recognition. MSE, R^2 and MAE were computed as performance metrics, confirming these visual results. It should be noted that due to the much higher average values of Price (that go up to, and over 1.000.000€), the MAE, RMSE and MSE will consequently be much higher when compared to Price/m^2, whose average values are several orders of magnitude inferior. R^2 is a more normalized metric in this case.

Table 2: Results for Predictions of Price and Price/m^2

	R^2	MSE	MAE	RMSE
Price	0.573	23999181815.733€^{2}	129720.789€	154916.693€
Price/m2	0.383	393554.840€$^2/m^4$	501.737€$/m^2$	627.339€$/m^2$

The coefficients for the model equations also show this: for Price/m^2, the higher Parish coefficients are coincident with the higher average Price/m^2 per Parish (e.g. the Parishes that have the higher coefficients for the model will be the same Parishes with the higher average Price/m^2). The same doesn't happen so noticeably with Price, showing that more variables are in play in order for an house in a certain location to be more expensive, than just the Parish where the house is located.

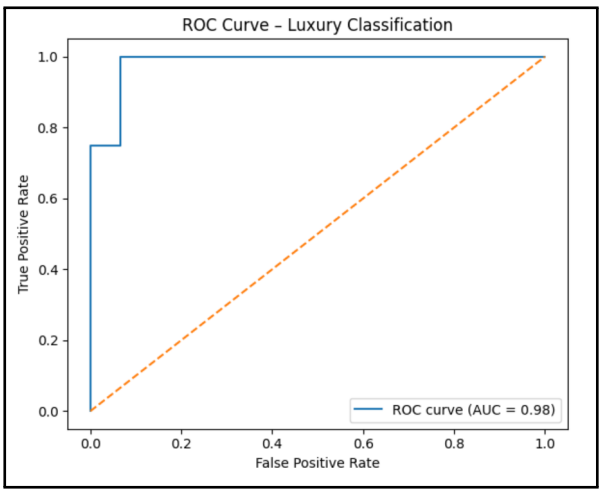
Moreover, these same models were constructed without the data exploration in mind (e.g. the models were created with PropertyType and PropertySubType in the dataset, even though these variables are clearly in an imbalance), and the results worsened (the metrics returned between 20-40% inferior values), emphasizing the model distortion previously mentioned. Particularly, for Price/m^2, when these two features are used, the aforementioned behavior for the Parish coefficients and their correlation with their average Price/m^2 becomes more pronounced. When these features are removed, some Parishes with lower average Price/m^2 start having higher contributions as coefficients (Benfica and Marvila are two good examples). This mostly contributes to the idea on how these two features can easily distort and confuse the model.

5 Classification

5.1 Model prediction to place a house in a market segment

A classification model was implemented to check if it's possible to place a house in a market segment simply by looking at its features, and to see if location can inflate these values.

Using as a target a new variable **luxurious**, true for any variable whose price surpasses 1.000.000€, and training a model based solely upon on the the characteristics that the house displays, the ROC AUC score was computed, as well as the coefficients of the model. The results are what follows:



(a) ROC curve for Luxury classification

	Feature	Coefficient
33	Parish_Santa Maria Maior	2.593430
27	Parish_Marvila	-1.143657
0	Bedrooms	-1.058112
1	Bathrooms	1.031913
11	PropertySubType_Apartment	-0.934866
14	Parish_Alcantara	0.870087
4	Parking	0.839806
26	Parish_Lumiar	-0.725892
30	Parish_Parque das Nacoes	-0.588959
10	Condition_Used	0.580557
8	Condition_For Refurbishment	-0.517924
23	Parish_Campolide	-0.473991
29	Parish_Olivais	-0.452866
20	Parish_Belem	0.359148
21	Parish_Benfica	-0.336814
9	Condition_New	-0.326376
18	Parish_Avenidas Novas	0.295841
22	Parish_Campo de Ourique	0.291292
7	Condition_As New	0.262573
5	Latitude	-0.224320
13	Parish_Ajuda	-0.214639
34	Parish_Santo Antonio	-0.089887
15	Parish_Alvalade	-0.089150
6	Longitude	-0.075367
25	Parish_Estrela	-0.073968
35	Parish_Sao Domingos de Benfica	-0.068061
17	Parish_Arroios	-0.041555
36	Parish_Sao Vicente	-0.040188
31	Parish_Penha de Franca	-0.031180
16	Parish_Areeiro	-0.025973
3	AreaGross	0.022175
2	AreaNet	0.011087
28	Parish_Misericordia	-0.006641
32	Parish_Santa Clara	-0.005535
12	PropertySubType_Studio	-0.003397
24	Parish_Carnide	-0.001441
19	Parish_Beato	-0.000571

(b) Coefficients obtained for the classification model

Figure 8: Prediction for the classification model

5.1.1 Model Evaluation

The model returned near perfect results (with an accuracy of 0.94), meaning that it can confidently place a house in a specific segment simply by observing the features of the houses. Location doesn't play such a significant role, as some other features, namely the type and condition of the house, also have a high contribution to the model, and can influence the pricing of the house, something that was already concluded with the Linear Regression model.

As for the Linear Regression, this model was also created using PropertyType and PropertySubType in the dataset, ignoring what the data exploration returned. The results remained mostly unchanged (only the accuracy was slightly diminished, but it's almost negligible), meaning that said variables don't particularly affect a classification approach, as much as they do when we try to predict price. The model can still place a house in the market without knowing what type or subtype of house it is. In other words, these two features have small weight as coefficients in the model.

5.1.2 Cross-Validation

To confirm the aforementioned classification results, a 5 fold stratified cross-validation was implemented, to check if the model is indeed strong. Taking into account the overall small dataset used, one simple split of the data could prove misleading and lead to model distortion.
5 splits were applied and the different ROC AUC scores computed, giving the below results:

$$\text{ROC AUC (5-fold CV): } [0.908, 1.000, 0.972, 0.950, 0.961]$$

Quite distinctly, it's possible to confirm that all of the results illustrate a very strong model, meaning that it can very confidently place a house in a luxury segment simply by observing the house features.

6 Decision Tree

6.1 Model with rules of the market

In order to educate people about market rules, a visual model with explicit and intuitive decision paths is highly effective, in this case the best choice is a decision tree model.

To demonstrate this concept, here is a tree with a maximum depth of 3 and a minimum of samples equal to 3 to prevent overfitting we obtain:

As shown, the results are easily interpretable. By applying a grid search to get the best hyperparameter, we obtain a maximum depth of 6 and a minimum of samples equal to 5.

6.2 Model Evaluation

For the tree with the best hyperparameter, the following results were obtained:

R^2	MAE (€)	MSE (€^{2})
0.727	87 383.52	15 361 873 888.48

Since the model is based on a decision tree, the results are far from optimal.

7 SVM classification

7.1 Houses prices band

In order to classify the price class (band) of an house it was need to divide the Price of the houses in 3 segments. With that, a new variable was defined for the model to predict.

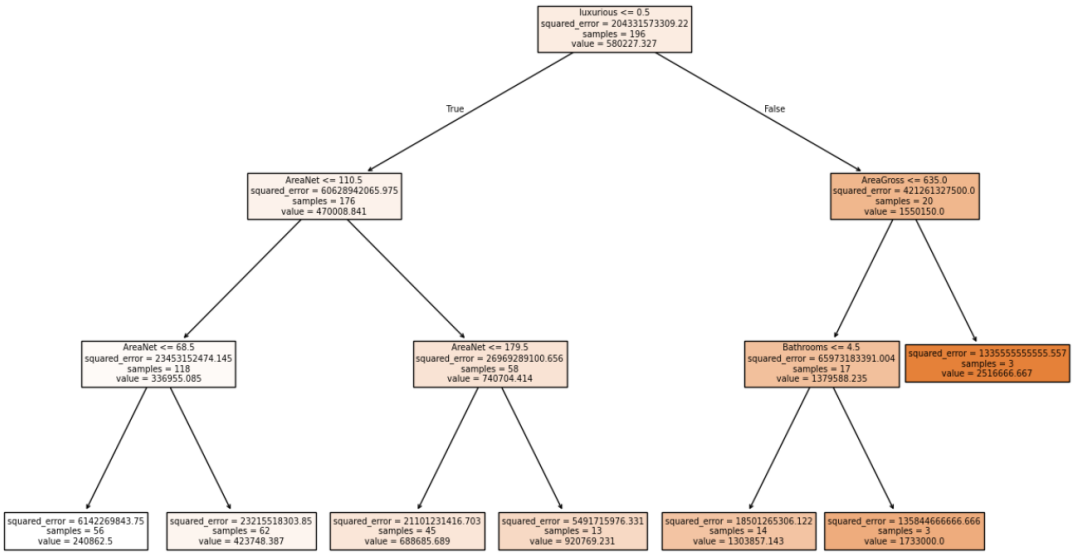


Figure 9: Simple Tree

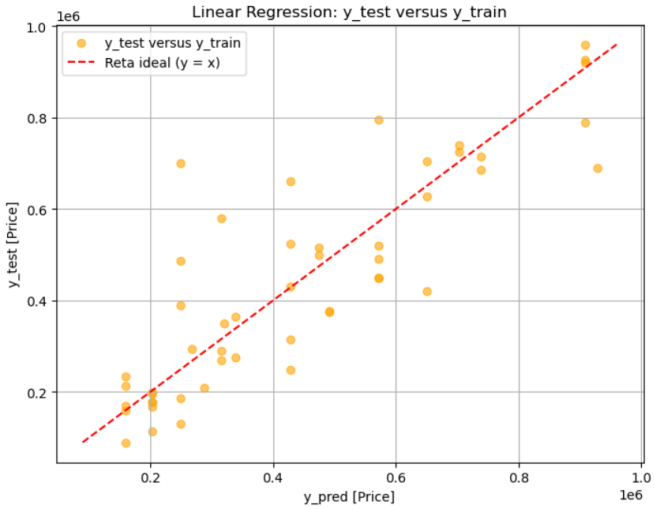


Figure 10: Real Price vs Predicted Price

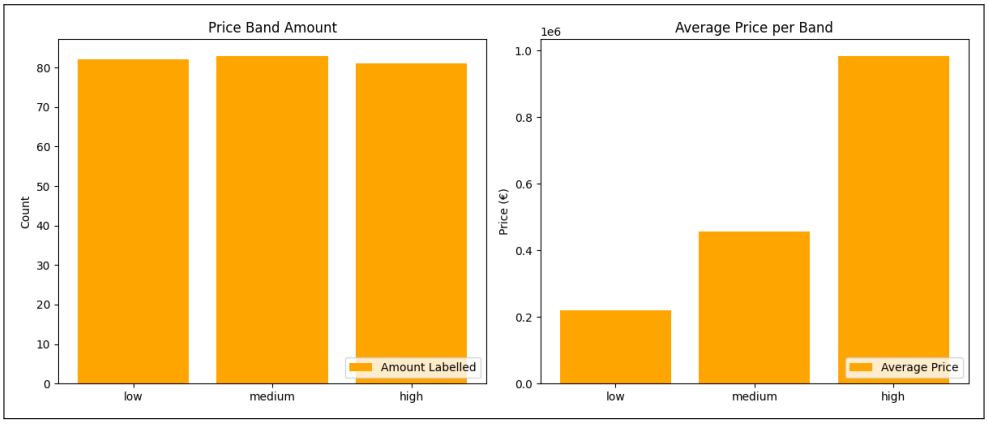


Figure 11: Price Band variable

7.2 Model to predict the price band

SVM was the best choice for the job due to its strong performance in high-dimensional feature spaces and its ability to handle non-linear decision boundaries through the use of kernel functions, *RBF*.

In order to find the best parameters for *RBF Kernel* a cross-validation method was applied, *GridSearch*. Using the method requires selecting a performance index. Three metrics were presented:

- Multi-variable ROC AUC;
- MAE;
- F1-Weighted.

Unfortunately, ROC AUC is not suitable for this case, because it penalizes equally the model mismatched results. It is expected to penalize more when the model gives low $\rightarrow$ high than low $\rightarrow$ medium. For that the last two are the best choice. To use MAE it is needed to change the variable for an ordinal score, for example, low = 0, medium=1, high=2, this makes sense since our variable is literally ordinary, $high > medium > low$. For F1-weighted this step is not needed.

RBF Kernel, for both MAE and F1-weighted, found $C = 40$ and $\gamma = 0.004$ as the best parameters that generalizes.

7.3 SVM results

MAE: 0 — *F1-weighted*: 0.77689

Table 3: Test Set (True values) vs Predictions (Model answers)

Index	True	Pred	Index	True	Pred	Index	True	Pred
1	medium	medium	11	low	low	21	high	high
2	medium	low	12	high	high	22	medium	low
3	high	medium	13	low	medium	23	high	high
4	high	high	14	high	high	24	low	low
5	high	high	15	medium	medium	25	low	low
6	low	low	16	medium	medium	26	low	low
7	medium	high	17	low	medium	27	high	high
8	high	high	18	low	low	28	high	high
9	low	low	19	medium	high	29	medium	medium
10	high	high	20	medium	medium	30	medium	medium

The model demonstrates excellent predictive performance on the test set, **Table 3**. There are no misclassifications between low and high, which represents the most critical type of error for this problem. The evaluation metrics (MAE and F1-weight) support this observation, both confirm the high accuracy of predictions, emphasizing that the model consistently performs well across all classes.

8 Conclusions

Having structured the project between data exploration and the construction of the models, it's now possible to grasp some conclusions based on the metrics obtained over the different objectives established.

For the Linear Regressions, the results were consistent with the first conclusions gathered from the data: **Price** as a feature shares correlations with the other features in the dataset, and as a result it's possible to predict house pricing with relative accuracy, even though these predictions become harder as the price of houses increase, due to the more pronounced outliers. **Price/m^2** is much more volatile in the other hand, since it doesn't share an equally linear pattern, with the Parishes being exceptions. The premise that some features like **PropertyType** and **PropertySubType** could prove to distort the models also turned out to be true, as the metrics pointed out.

The Classification model, on the other hand, displayed excellent results, with a near-perfect accuracy, a sign that the housing characteristics separate price segments much better than they explain prices. Luxury housing has such strong features that the model doesn't find it hard to generalize it and place in a specific segment. Regression suffers more from "noise", in the sense that in real estate more variables are in play to explain prices (renovation quality, seller urgency, market timing, angle facing the sun, etc...), variables that don't exist in this dataset. Classification ignores this "noise", as the few features that the dataset has are sufficient to place a house in a specific segment.

Regarding the decision tree model, it offers a clear and interpretable insights into the market, making it simpler to understand, but even after the hyperparameter optimization, its predictive performance was very limited. So despite its transparency, this model is not robust enough.

The SVM model shows excellent predictive performance. Importantly, there are no misclassifications between the low and high classes, confirming the model's reliability and overall accuracy on classifying house price categories. This indicates that the model can effectively assist in identifying over- or under-priced properties and supports informed decision-making in the housing market.